

Lab 1 - TypeScript & NestJS Blog API

Welcome! 🎉

In this lab you'll practice TypeScript basics and build a simple Blog API using NestJS and MongoDB.

Part 1 - TypeScript (10 Questions)

- Q1.** What's the difference between let, const and var?
- Q2.** What's the difference between interface and type? Give one example for each.
- Q3.** Create an interface for a User object containing: name, email, age and isAdmin.
- Q4.** Write a function that receives two numbers and returns their sum.
- Q5.** Write a function that accepts string | number. If it's a string, return its length. If it's a number, return its square.
- Q6.** Create a generic function called firstElement() that returns the first element in any array.
- Q7.** Create a User class with name and email and a printInfo() method.
- Q8.** What's the difference between any and unknown?
- Q9.** Create an enum called Role with: Admin, User and SuperAdmin.
- Q10.** Explain the difference between == and === with one example.

Part 2 - Build a Blog API 🚀

Technology:

- NestJS
- MongoDB
- Mongoose
- TypeScript

Schemas

User:

- name
- email
- password
- age
- image

Post:

- title
- content
- image
- author (ObjectId)

Relationship: One User → Many Posts (1 : M)

Your Tasks

- Create full CRUD APIs for User.
- Create full CRUD APIs for Post.
- Before creating a post, make sure the user exists.
- Save the user's ObjectId in the author field.
- When deleting a user, delete all posts that belong to that user.
- Populate author information when getting posts.
- Email must be unique.
- Hash the password before saving.
- Validate requests using ValidationPipe and class-validator.
- Use .env to store PORT and MONGO_URI.
- Organize the project using NestJS Modules.

Bonus

- Pagination
- Search by Title
- Sort Posts
- add .env
- File Upload